# FINITE PROJECTIVITY OVER A GLUEING PAIR IS DETECTED BY THE TWO PIECES OF THE PAIR

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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*Editorial note.* The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

*Editorial note.* The author refers to 'the lemma above' at proof line 25799 for the statement established earlier in the section which is counted by the card S15-alg-BL-flat-criterion; that is a navigation pointer and no part of its counted proof span falls inside this card. The four references inside the counted proof span are recorded as non-load-bearing navigation pointers.

*Allowed standard prerequisites (editorial).*

- The flatness criterion for a glueing pair, by which flatness of M follows from flatness of its two pieces. invoked at proof line 25792. Is the first of the two ingredients feeding into the finite-projective criterion.
- Faithful descent of finiteness along a glueing pair. invoked at proof line 25793. Gives finiteness of M from the two pieces.
- The criterion characterising finite projective modules via finite presentation, and exactness of tensoring a short exact sequence with R' and R_f. used at proof lines 25790-25799. Reduces finite projectivity of M to finite generation of the kernel K.

*Source reference.* Original lemma 00NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 875–875. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 00HL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 647–647. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 25708: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25708–25708. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

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*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 25190: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25190–25190. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

**Lemma 0.2.** *Let $(R \to R', f)$ be a glueing pair. Let $M$ be an $R$-module which is not necessarily glueable for $(R \to R', f)$. Then $M$ is a finite projective $R$-module if and only if $M \otimes_R R'$ is finite projective over $R'$ and $M_f$ is finite projective over $R_f$.*

*Proof.* Assume that $M \otimes_R R'$ is a finite projective module over $R'$ and that $M_f$ is a finite projective module over $R_f$. Our task is to prove that $M$ is finite projective over $R$. We will use Algebra, Lemma 00NX without further mention. By Lemma 25708 we see that $M$ is flat. By Lemma 25190 we see that $M$ is finite. Choose a short exact sequence $0 \to K \to R^{\oplus n} \to M \to 0$. Since a finite projective module is of finite presentation and since the sequence remains exact after tensoring with $R'$ (by Algebra, Lemma 00HL) and $R_f$, we conclude that $K \otimes_R R'$ and $K_f$ are finite modules. Using the lemma above we conclude that $K$ is finitely generated. Hence $M$ is finitely presented and hence finite projective. $\square$